

Ezechinyere Ukabiala

PhD Candidate at UC Berkeley - UCSF Bioengineering

An extremely motivated PhD student with about 8 years of research experience spanning several bioengineering fields. I have led various projects and presented at various academic conferences. I am detail-oriented and a problem solver.

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EDUCATION

Doctor of Philosophy: Bioengineering

UC Berkeley/UC San Francisco

09/2022 - Present

Bachelor of Science : Bioengineering (Biotechnology)

UC San Diego

08/2016 - 09/2020

TEACHING EXPERIENCE

BENG 1 (Introduction to Bioengineering)

University of California San Diego

01/2020 - 03/2020

BE-BOAT (Bioengineering Broad Overview and Techniques) Research Program

University of California San Diego

06/2019

PUBLICATIONS

Research paper at peer-reviewed Journal

Low-N Protein Activity Optimization with FolDE

Author(s)

Co-author

28/10/2025

In review

Research paper at peer-reviewed Journal

Polyketide synthase-based controlled synthesis of polycyclopropanated fuel molecules

Author(s)

Co-author

20/02/2025

Accepted (Nature communications)

ORGANIZATIONS

Bioengineering Association of Students at UC Berkeley & UCSF (12/2025 - Present)

President

WORK EXPERIENCE

Graduate Student Researcher

Keasling Laboratory

06/2023 - Present

Achievements/Tasks

- Cloned and chromosomally integrated the *pop* biosynthetic gene cluster from *S. roseovirens* to *S. venezuelae*, confirming cyclopropanated fatty acid production using LC-MS.
- Engineered overexpression constructs pyruvate transaminase (*bauA*) and malonyl-CoA reductase (*mcr-c*) to increase extender-unit supply.
- Aided in designing and building chimeric POP-PKS variants to generate distinct classes of polycyclopropanated fatty acids.
- Developed and validated a GC-MS-based workflow for the detection of alkenes and fatty acids to low micromolar levels.
- Executed purified-enzyme reactions with OleT-PRN on C12, C12 and C16 straight chained fatty acids confirming terminal-olefin formation.

Research Associate

Alivamab Biologics

09/2021 - 06/2022

Achievements/Tasks

- Aided in the discovery of therapeutic monoclonal antibodies and using Hybridoma technology.
- Aided in the development of bispecific antibodies using Hybridoma technology.
- Carried out high throughput screening of hybridomas using FACS and ELISA.
- Carried out assay development and optimization.
- Assessed antibody binding affinity (EC50) using FACS.
- Determined binding rates of antibodies using Bio-Layer Interferometry (Octet and Gator).
- Performed aseptic cell culture activities including cell culture maintenance and cell counting.
- Carried out general lab maintenance including cleaning, restocking and organizing lab supplies.

ORGANIZATIONS

African Student Association at UC San Diego
(06/2018 - 06/2019)
Vice President

African Student Association at UC San Diego
(06/2017 - 06/2018)
Secretary

HONOR AWARDS

Chancellor's Scholar (08/2022 - 05/2023)
University of California Berkeley

McNair Scholar (10/2018 - 08/2019)
University of California San Diego

Excellence in Presentation of Research at the 32nd Annual Undergraduate Research Conference (05/2019)
University of California San Diego

CAMP (California Louis Stokes Alliance for Minority Participation) Scholar (10/2018 - 08/2019)
University of California San Diego

Provost Honors (01/2017 - 03/2018)
University of California San Diego

Provost Honors (01/2019 - 06/2019)
University of California San Diego

Provost Honors (09/2019 - 03/2020)
University of California San Diego

CONFERENCES & COURSES

Orthopaedic Research Society (ORS) 2020 Annual Meeting
Orthopaedic Research Society (ORS)

- *Multimodal and Multiscale 3-D Imaging of Postnatal Skeletal Growth of The Murine Femoral Head*

Undergraduate Research Conference (URC) 2019
University of California San Diego

- *Multimodal and Multiscale Imaging of Murine Skeletal Growth : Hip and Spine*

Summer Research Conference (SRC) 2019
University of California San Diego

- *Multimodal and Multiscale Imaging of Murine Spine Growth*

Bioengineering Day 2020
University of California San Diego

- *Multi-scale Mapping of Skeletal Growth Biomechanics and Biology*

WORK EXPERIENCE

Laboratory Technician Truvian Sciences

02/2021 - 09/2021

Achievements/Tasks

- Aided teams of scientists in assay optimization and troubleshooting.
- Aided teams of engineers in product development and optimization.
- Conducted high throughput clinical chemistry assays, hematology assays and immunoassay.
- Performed optical analysis on stained blood cells using light and fluorescence microscopy.
- Created SOPs for laboratory techniques and procedures.

Undergraduate Student Researcher Cartilage Tissue Engineering Laboratory (Sah Laboratory)

05/2018 - 09/2020

Achievements/Tasks

- Aided in the definition of a multiscale framework for cellular and matrix-bases of murine vertebral growth pertaining to Adolescent Idiopathic Scoliosis and Degenerative Disk Disease.
- Carried out histological analysis using 2D and 3D histological block face techniques and image analysis with light, fluorescence .
- Created SOPs for new laboratory techniques and procedures.
- Analyzed large databases of information using Microsoft Excel and Word.
- Carried out general lab maintenance including cleaning, restocking and organizing lab supplies.

Undergraduate Student Researcher Kwon Laboratory

10/2017 - 03/2018

Achievements/Tasks

- Developed aseptic tissue cultures activities of Neuro 2A cells including cryopreservation and cell counting.
- Analysed proteins via BSA assays, SDS-PAGE, Western blotting.
- Carried out gel electrophoresis and nanoparticle creation.
- Integrated, improved and analyses large databases with Microsoft Excel.
- Carried out general lab maintenance including cleaning, restocking and organizing lab supplies.